AC Battery vs. Two-Level Inverter: Map and Drive-Cycle Efficiency from Passenger Cars to Trains

H01 Power Electronics for E-Mobility
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Abstract

Battery-integrated modular multilevel converters (BI-MMCs), or AC batteries, replace the traction inverter with battery submodules switched by low-voltage Si MOSFETs. We present an open-source platform that compares the efficiency maps of a BI-MMC and a two-level SiC inverter, including MTPA operation and the battery, and evaluates their drive-cycle efficiency with SOC and temperature tracking. For six applications from cars to trains, the BI-MMC cuts semiconductor losses by 26 % to 80 %, but its cells carry the phase current: it gains only about 0.1 pp where all energy is below base speed and loses up to 1.7 pp elsewhere.

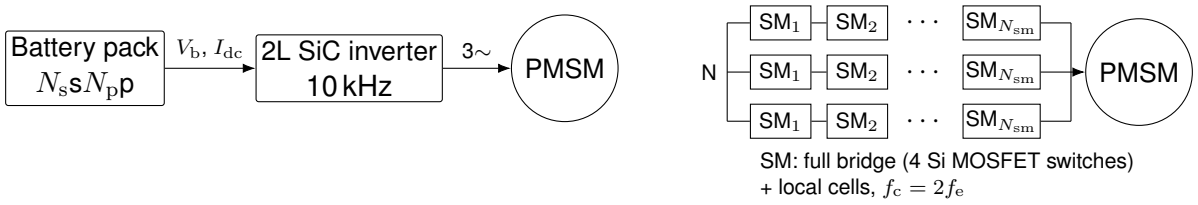
1 Synopsis

In a conventional battery-electric drive, the pack is a single high-voltage string feeding a two-level (2L) SiC inverter that switches the full DC-link voltage at a fixed carrier frequency. The battery-integrated modular multilevel converter (BI-MMC), also known as an AC battery or modular multilevel battery [1], [2], [3], instead distributes the cells over N_{sm} series-connected full-bridge submodules (SMs) per phase. Each SM inserts, bypasses or reverses its local cell string, so the phase voltage is synthesised directly from the batteries. This enables low-voltage Si MOSFETs with very low on-resistance, SM switching at a few times the fundamental frequency, and inherent cell balancing. The price is that the cells carry the switched phase current, including its reactive and field-weakening components, which raises the battery Joule losses [4].

Whether the BI-MMC is the more efficient drive therefore depends on where in the torque–speed plane a vehicle spends its energy. Published comparisons consider few operating points or one vehicle, or neglect the battery. This paper contributes: (i) an open-source evaluation platform that compares the efficiency maps of the two-level inverter and the AC battery over the full torque–speed plane, with identical machine, cells and installed energy, including the battery losses; (ii) a drive-cycle evaluation in the same platform, which steps through a duty cycle, tracks SOC and temperatures, and returns the cycle efficiency of both drives; (iii) a study of six applications, from passenger cars to off-road machines, ships and trains, which yields a simple rule for when an AC battery pays off.

2 Evaluation Platform

Figure 1 shows the two architectures. Each application in table 1 uses an interior PMSM ($p = 4$) with the same per-unit parameters, the same 24 A h NMC cell, the Wolfspeed EDB005M12TM4 SiC module [5] with measured, current-dependent switching slopes and temperature-dependent $R_{\mathrm{ds,on}}$ for the 2L inverter, and 0.3 mΩ Si MOSFETs (four in parallel) switching at $f_{\mathrm{c}} = 2f_{\mathrm{e}}$ in the BI-MMC. The cells per SM, $N_{\mathrm{s,sm}}$, are an input; $N_{\mathrm{sm}} = \lceil V_{\max}/(V_{\mathrm{nom}}N_{\mathrm{s,sm}})\rceil$ lets the BI-MMC reach the 2L phase voltage, with the same energy in both drives.



(a) (b)

Fig. 1: Compared architectures: (a) battery pack with a two-level SiC inverter; (b) BI-MMC with three star-connected strings of N_sm battery-integrated full-bridge submodules.

Loss maps. For every point of a 50×100 speed–torque grid covering motoring and regeneration, the stator currents follow from maximum-torque-per-ampere (MTPA) optimisation, solved with CasADi/IPOPT [6] separately for each drive's voltage limit:

$$\min_{i_d, i_q} \; i_d^2 + i_q^2 \quad \text{s.t.} \quad \tfrac{3}{2} p\big(\psi_\mathrm{f} i_q + (L_d - L_q) i_d i_q\big) = \tau_\mathrm{e}^*, \qquad v_d^2 + v_q^2 \le \hat{v}_\mathrm{ph,max}^2(z), \tag{1}$$

where $\hat{v}_\mathrm{ph,max}$ follows the cell open-circuit voltage at SOC z: $v_\mathrm{oc} N_\mathrm{s}/\sqrt{3}$ for the 2L inverter and $v_\mathrm{oc} N_\mathrm{s,sm} N_\mathrm{sm}$ for the BI-MMC. With peak current $\hat{\imath}$ and N_p parallel dies, the conduction losses are $\frac{3}{2} R_\mathrm{ds,on}(T_\mathrm{j}) \hat{\imath}^2 / N_\mathrm{p}$ (2L, iterated with the thermal network) and $3 N_\mathrm{sm} R_\mathrm{ds,on} \hat{\imath}^2 / N_\mathrm{p}$ (BI-MMC, two switches conduct per SM). Switching energies follow the slope-based method of [7], $E_\mathrm{sw} = \frac{1}{2} V \bar{\imath}\,(t_\mathrm{ri} + t_\mathrm{fv} + t_\mathrm{fi} + t_\mathrm{rv})$, with V the DC-link or SM voltage. The cells are SOC-dependent first-order equivalent circuits. The 2L pack carries only the active-power current I_dc; in the BI-MMC, the cell current is the phase current gated by the unipolar PWM state $s(t) \in \{-1, 0, 1\}$ and simulated in the time domain:

$$i_\mathrm{b}(t) = s(t)\, i_\mathrm{ph}(t), \qquad \frac{\mathrm{d}v_C}{\mathrm{d}t} = \frac{1}{C_1}\Big(i_\mathrm{b} - \frac{v_C}{R_1}\Big), \qquad P_\mathrm{b} = \big\langle R_0 i_\mathrm{b}^2 + v_C^2 / R_1 \big\rangle_{T_\mathrm{e}}. \tag{2}$$

Duty cycles. Road and rail speed profiles are converted to machine speed and torque with a road-load model $F = k_\mathrm{m} m \dot{v} + \frac{1}{2} \rho c_\mathrm{d} A v^2 + c_\mathrm{rr} m g$ (Davis resistance for the train), the wheel radius and a gear ratio. The truck (40 t) drives the 119 km Södertälje–Norrköping road with its measured slope profile, the off-road machine follows the NRTC, and the ship the four propeller-law modes of ISO 8178 E3. P_b and the gear ratio are sized so that each cycle stays within the torque and power limits. Instead of interpolating the maps, the evaluator steps through each cycle: at every sample it solves Eq. (1) at the present SOC, evaluates the loss models of both drives, updates SOC from the charge drawn per cell and evaluates a steady-state thermal network for the junctions, cases and cells. Torque beyond the common envelope of both drives is assigned to the mechanical brake. The cycle efficiency is $\eta_\mathrm{cyc} = 1 - E_\mathrm{loss}/(E_\mathrm{mot} + E_\mathrm{reg})$.

Tab. 1: Applications, duty cycles and cycle efficiency at 50 % SOC (semiconductors and battery). $\Delta\eta = \eta_\mathrm{BI} - \eta_\mathrm{2L}$; the range in brackets covers 20 % to 90 % SOC.

Application	P_b [kW]	V_dc [V]	E_b [kW h]	n_b [rpm]	$N_\mathrm{sm} \times N_\mathrm{s,sm}$	Duty cycle	η_2L [%]	η_BI [%]	$\Delta\eta$ [pp]
Passenger car	150	400	75	5000	9×7	WLTC Class 3b [8]	98.87	98.38	-0.49 [-0.77, -0.34]
City bus	350	700	350	2500	11×10	Braunschweig [9]	98.35	97.78	-0.57 [-0.86, -0.48]
Heavy-duty truck	550	700	400	3000	11×10	Södertälje–Norrköping	98.45	97.62	-0.83 [-1.58, -0.48]
Off-road machine	150	700	200	2000	11×10	NRTC [10]	99.02	99.10	$+0.09$ [$+0.02$, $+0.09$]
Marine vessel	400	800	1000	1800	13×10	ISO 8178 E3 [11]	99.06	99.17	$+0.10$ [$+0.05$, $+0.11$]
Regional train	600	800	300	2000	13×10	Synthetic, 3 stops	97.15	95.40	-1.74 [-2.34, -1.68]

3 Results

Figure 2 contrasts two applications. Below base speed, the BI-MMC is more efficient over most of the map where the 2L inverter's 10 kHz switching and conduction losses dominate (everywhere below base speed in the off-road machine, locally by up to 6 pp). Above base speed, the field-weakening current flows through every inserted cell, and the BI-MMC loses 0.1 pp (off-road) to 1.1 pp (truck) on average. The off-road machine spends 79 % of its cycle energy below base speed and gains 0.09 pp; the truck spends all of it above base speed and loses 0.83 pp.

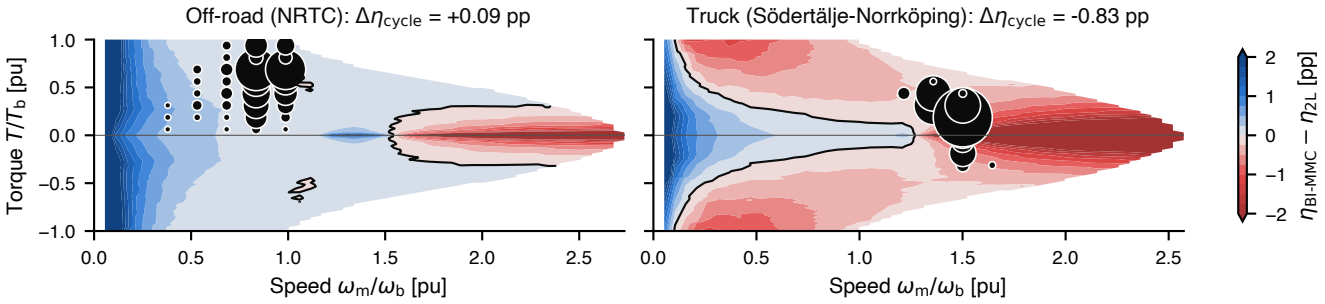


Fig. 2: Efficiency difference BI-MMC − 2L (semiconductors and battery) over the torque–speed map at 50 % SOC, for the off-road machine and the heavy-duty truck. Solid line: $\Delta\eta = 0$. Dots: cycle energy $|P|\Delta t$ per bin; dot area is proportional to the share of the cycle energy.

Figure 3(a) breaks the cycle losses down. In all applications the BI-MMC cuts semiconductor losses, by 37 % to 80 % in road, off-road and marine duty and by 26 % in the train, where 13 series SMs carry the high phase current; switching losses almost vanish and peak junction temperatures drop by 6 K to 25 K. In return, its battery losses are 1.6 to 4.4 times higher. Figure 3(b) condenses this into a design rule: the BI-MMC only gains when nearly all cycle energy is delivered below base speed at moderate to high torque (ship, off-road machine), and even then the gain is about 0.1 pp. The passenger car and the city bus lose 0.5 pp to 0.6 pp: the car has 62 % of its energy below base speed, but its light WLTC loads keep the 2L losses low. The truck (–0.83 pp) and the regional train (–1.7 pp) spend most of the cycle in field weakening. Low SOC deepens field weakening and penalises the BI-MMC; the off-road gain nearly vanishes at 20 % SOC.

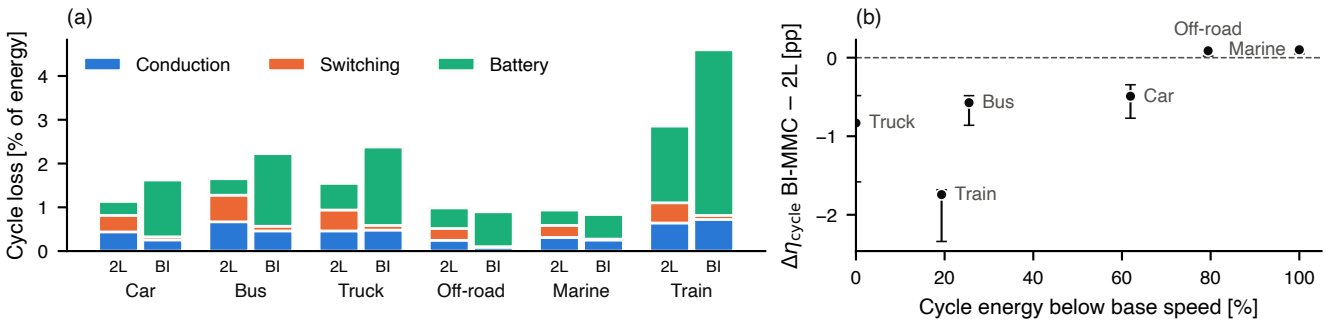


Fig. 3: (a) Cycle losses by component in percent of the traction energy throughput at 50 % SOC; (b) cycle efficiency difference BI-MMC − 2L versus the share of cycle energy below base speed (bars: 20 % to 90 % SOC).

4 Conclusion and Content of the Final Paper

The open-source platform puts the efficiency maps and drive-cycle efficiency of an AC battery and a 2L inverter on equal terms. Across six applications, the AC battery cuts semiconductor losses, but the cell losses from the gated phase current offset this: it is at best on par with a SiC 2L inverter at low speed and clearly worse in field weakening (trucks, trains). The platform will be released as open source; the final paper adds a sensitivity study.

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